

[DC] Precise Hand Tracking using Multiple Optical Sensors

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ABSTRACT

Virtual and extended reality are some of the fastest-growing fields today. The visual quality, the headset size, the latency – it all improves with every new device. However, the user interface is still falling behind.

After several decades, when we mainly used a mouse and keyboard as controllers for personal computers, we finally can and should use other methods of input. We can even use parts of our body other than our hands to control the virtual worlds. For example, we can track the eyes for additional input or run around the virtual world with our own feet.

This doctoral thesis deals with hand-tracking controllers for virtual reality without the user needing to hold or wear any device to interact with the virtual world. The goal is to create an intuitive controller using optical sensors that can provide a more natural user interface.

We propose and implement a set of algorithms to fuse the tracking data from multiple optical hand-tracking sensors that provide greater precision and tracking possibilities than a single optical sensor.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

In recent years, many research articles have already focused on creating an intuitive user interface device for virtual reality. The inability to control the virtual environment naturally just by moving our hands without the need for any controller is a big motivation to proceed in this field. By using multiple optical sensors, we would be able to provide a precise and intuitive user interface that allows the user to move freely and interact easily with the virtual world.

During the analysis of the optical hand-tracking sensors and of state-of-the-art research in this field, we identified the main flaws and drawbacks of optical hand-tracking, like short tracking range, small field of view, occlusion of hands and fingers, and chirality misclassification.

In this thesis, we combine the tracking data from multiple already existing hand-tracking sensors to overcome the flaws and limitations of the optical hand-tracking approach. We introduced and implemented a set of algorithms to create a precise hand-tracking system that outperforms classic hand-tracking with one sensor. We focus on synchronizing and fusing the hand-tracking data streams to achieve more precise tracking data. The proposed approach is displayed on a scheme in Fig. 1.

2 SYSTEM FOR A PRECISE HAND-TRACKING

During the work on this thesis, we presented a set of algorithms to fuse data from multiple optical hand-tracking sensors [4], later extended in [3]. Even though we use the Ultraleap hand-tracking

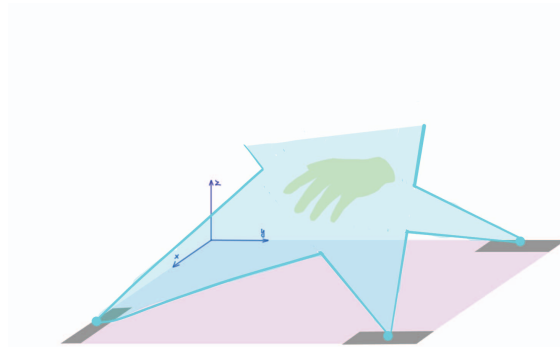


Figure 1: Proposal of multiple optical hand-tracking sensor approach.

sensor as a representative of the technology, the proposed algorithms are universal and do not depend on any specific sensor. Our approach fixes the short tracking range, small field of view, occlusion of the hand and fingers, and hand chirality misclassification and generally provides more precise hand tracking.

2.1 Ultraleap sensor

Ultraleap Stereo IR 170 sensor by Ultraleap [6] is the successor of the Leap Motion Controller [5] and one of the most successful hand-tracking sensors on the market. The sensor uses infrared light to illuminate the hand and then tracks the reflection of the infrared waves by two built-in cameras. The images from the sensor are processed by undisclosed (proprietary) classical vision techniques, which resolve the position of hands and fingers.

The hand-tracking data are then provided via API as structured information about 27 distinct elements of the hand, such as the position and rotation of the palm, fingers and bones. The information about each finger is further divided into individual units according to bones and joints.

If part of the hand is occluded or out of the sensor's field of view, the tracking software estimates the tracking data for the missing part according to the visible part. The tracking software also provides interpolation and extrapolation of the tracking data to provide smoother tracking.

2.2 Fusing data from multiple sensors

In the doctoral thesis, we proposed and tested an approach for using multiple optical sensors to provide more precise and robust hand tracking. First, we focused on finding a way to fuse the hand-tracking data from multiple optical sensors [4]. Later, we extended this fusing algorithm and made the algorithm for synchronising the sensors easier and faster to use [3]. After that, we compared the Kabsch algorithm [2] and the Iterative closest point algorithm [1] for synchronizing the hand-tracking data in space and improving the computation of hand-tracking data confidence. Our approach allows

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us to extend the tracked space and make the tracking more precise by adding more sensors.

We are proposing the following four main algorithms in our doctoral thesis:

space calibration algorithm is used to synchronize the hand-tracking data from multiple optical sensors so it can be fused,

hand-tracking data confidence algorithm determines which sensor tracks the hand the best at a given time,

hand-tracking data fusion algorithm processes the tracking data from all the connected sensors and fuses them into one output,

hand chirality confirmation determines which sensor reports the correct hand chirality.

Combining these four approaches allows us to provide precise hand tracking by tracking the hand from multiple angles with multiple sensors.

2.2.1 Space calibration algorithm

The space calibration algorithm, presented in [4] and later extended in [3], synchronizes the data streams from all available hand-tracking sensors. The goal is to find and apply the correct transformation (translation and rotation) on the tracking data from all sensors so they are in absolute coordinates, not relative.

Initially, one sensor, called the pivot, is marked as calibrated. Then, for every time t , the calibration algorithm stores **space calibration samples** – the hand-tracking data from all of the sensors that see the hand, along with the corresponding hand-tracking data from the pivot. When a predefined number of hand-tracking data for every sensor is stored, the correct translation and rotation are computed.

The rationale of this algorithm is very simple – the hand-tracking data from all of the sensors for time t for hand h are the same, just in a coordinate system relative to the corresponding sensor. Since the hand-tracking data for time t for sensor s_1 is just an affine transformation of the hand-tracking data from sensor s_2 , we can then compute the translation and rotation of the tracking data by estimating the transformation by point-cloud matching of data representing the hand reported by the sensors.

Currently, we tried the Kabsch algorithm [2] and the Iterative Closest Point algorithm [1] for the point-cloud matching and discovered that ICP is more precise but is much slower than Kabsch.

2.2.2 Hand-tracking data confidence

Our second algorithm for precise hand-tracking with multiple optical sensors is the hand-tracking data confidence algorithm [3]. It determines which sensor tracks the hand better at a given time.

This confidence value depends on both the angle of the hand to the sensor and the distance between them. In general, the sensor that sees the palm from the bottom can determine more information about the hand and thus has a higher hand-tracking data confidence than the sensor that sees the hand from the side (and probably sees the hand and fingers at least partly occluded). A sensor that is too close or too far from the hand will be assigned a lower hand-tracking data confidence value than a sensor that is at the optimal distance from the hand. The hand-tracking data confidence value is recomputed in real-time with every hand-tracking data sent by the sensor, so it always corresponds to the current hand pose.

2.2.3 Hand-tracking data fusion

The third algorithm we proposed in our thesis was the hand-tracking data fusion algorithm [4]. It builds up on top of the previous two algorithms – it takes the tracking data from the synchronized sensors and their hand-tracking data confidence to compute a weighted

average of the tracking data from all of the sensors. Since all the sensors provide some information about the hand, but some sensors have more precise data for a given time, we can determine which sensor affects the resulting tracking data the most. Since the hand-tracking data confidence changes with every hand-tracking data update, the weights used for the average computation must also change dynamically with every update.

2.2.4 Hand chirality confirmation

Since the hand chirality is often misclassified due to the similarity of the right hand to the left hand (and vice versa) when one of the hands is palm up, we proposed an algorithm that deals with it. In general, when more sensors report different chirality for the tracked hand than the rest, we use the chirality that is the same for the majority of reported hand-tracking data. When there is a tie, we use the information from the sensors with the highest sum of confidence values.

3 CONCLUSION

This thesis deals with precise hand-tracking with optical sensors, which can be used as a main interaction technique for virtual and extended reality.

In our research of the current state-of-the-art optical hand-tracking controllers, we identified the short tracking range, small field of view, occlusion of hands and fingers and common misclassification as the main problems. In the subsequent analysis, we discovered that no research or piece of technology successfully deals with these drawbacks or that they only focus on a subset of the said problems.

We proposed a series of algorithms that combine the hand-tracking data from multiple optical sensors, thus overcoming all of the said limitations of tracking with a single optical hand-tracking sensor, and we implemented and verified the proposed algorithms in a series of tests that compared our approach to state-of-the-art research and to the precision of a single sensor.

Areas for improvement:

1. How to improve the space calibration precision?
2. How to determine the best placement of the sensors?
3. How to provide haptic feedback?

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REFERENCES

- [1] A. Geiger, P. Lenz, and R. Urtasun. Are we ready for Autonomous Driving? The KITTI Vision Benchmark Suite. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3354–3361, 2012. doi: 10.1109/CVPR.2012.6248074
- [2] W. Kabsch. A solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 32(5):922–923, Sep 1976. doi: 10.1107/S0567739476001873
- [3] T. Novacek and M. Jirina. Project Multileap: Making Multiple Hand Tracking Sensors to Act Like One. In *Proceedings of 2021 IEEE 4th International Conference on Artificial Intelligence and Virtual Reality (AIVR 2021)*, AIVR 2021, pp. 19–26. IEEE, Taichung, Taiwan, 2021.
- [4] T. Novacek, C. Martin, and M. Jirina. Project Multileap: Fusing Data from Multiple Leap Motion Sensors. In *Proceedings of 2021 IEEE 7th International Conference on Virtual Reality (ICVR 2021)*, ICVR 2021, pp. 19–26. IEEE, New York, NY, USA, 2021.
- [5] Ultraleap. Leap Motion Controller, Oct. 2022. Accessed 31 October 2022.
- [6] Ultraleap. Ultraleap Stereo IR 170, Nov. 2023. Accessed 22 November 2023.